PHASE 2 · PROGRESS REVIEW

Sambodh IAS

A self-paced, AI-powered preparation platform for India's UPSC Civil Services Examination.

AI FOR IMPACT

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TARGET USERS AND CONTEXT

Two in three serious aspirants stall in "half-preparation"

6Lakh

serious aspirants attempt UPSC CSE each year, chasing roughly **1,000 vacancies**.

4Lakh

are self-driven but get **stuck mid-preparation** — knowing the syllabus, not closing the gaps.

₹1–2Lakh

typical annual coaching cost — out of reach for most Tier-2/3 self-starters.

Unstructured syllabus

A vast, sprawling syllabus with no map of what is mastered and what is missing.

No personalized feedback

Generic coaching can't tell one aspirant's weak concept from another's.

Isolation and burnout

Solitary, multi-year preparation drives high dropout and lost effort.

TARGET USERS AND CONTEXT

Who we build for

Primary

The repeat aspirant

Has sat Prelims or Mains at least once. Knows the syllabus cold — needs **targeted gap-filling**, not ground-up coaching. Studies self-paced, on a budget.

Secondary

The first-time serious aspirant

Building structured coverage from scratch and using the platform for honest **self-assessment** across the full syllabus.

Context

Tier-2 & Tier-3 India

Highly aspirational, entry-level Android phones, intermittent connectivity, and a strong preference to study in their **own language**.

SOLUTION OVERVIEW

A self-paced learning ecosystem with memory

Sambodh IAS turns an aspirant's own study material into adaptive assessments, grounded AI evaluation, and a study loop that remembers their journey.

Understands the journey

Tracks progress, strengths, and knowledge gaps across the entire syllabus — down to individual concepts and book pages.

Personalized assessment

Adaptive question selection driven by mistake patterns, with source-cited feedback on exactly what was wrong and why.

Support and mentorship

Revision nudges, a study companion, and pathways to peer and senior mentorship — countering isolation.

The core loop: assess, close the gap, reassess

Create assessment

Pick subject & topic; choose adaptive or seed, difficulty, PYQs.

→

Take it

Answer RAG-generated questions from real textbook content.

→

Get evaluated

Source-cited feedback: score, strengths, weaknesses, concept gaps.

→

Revise & reassess

SM-2 spaced-repetition queue resurfaces weak concepts when they're due.

Every attempt updates concept mastery, syllabus coverage, and the revision schedule — so the next assessment is sharper than the last. **The platform compounds what it knows about each aspirant.**

WHY AI IS NECESSARY

What fails without AI

Rules & static banks

- ✕ Fixed question banks can't generate fresh items from *this* aspirant's weak chunks.
- ✕ Keyword grading can't read a Mains answer for *Discuss / Analyze / Evaluate*.
- ✕ One study plan for everyone — no notion of individual mistake patterns.
- ✕ No way to tell a careless slip from a real conceptual gap.

Memory-augmented AI

- ✓ RAG generates questions grounded in the exact pages a learner is weak on.
- ✓ LLM evaluation scores subjective answers against directive verbs and rubrics.
- ✓ Adaptive engine reweights coverage from each learner's live mastery state.
- ✓ Mistake classifier separates factual, conceptual, and time-pressure errors.

📚 Factual gaps

→ flashcards & mnemonics linked to source

💡 Conceptual

→ explainers, analogies, worked examples

⏱ Time pressure

→ timed drills & strategy coaching

PHASE 2 · PROGRESS

What we planned — and what we shipped

COMMITTED IN PHASE 1	SHIPPED IN PHASE 2
Mains subjective evaluation + OCR for handwriting	✓ **5-dimension Mains rubric** + credit-beyond-reference fact-check (RAG/Exa); **Gemini-2.0-flash OCR** → GPT-4o for handwritten answers
Index Mains GS1–GS4 content	✓ **287 Mains questions** with structured Intro / Body / Conclusion model answers
Hindi + regional languages, end-to-end	✓ **8 Indian languages** (hi·te·ta·or·bn·mr·pa·kn) — translate-on-read + cache, term-preserving; built & verified
Feedback-quality loop & analytics	✓ **24,191 LLM calls instrumented** (model · latency · tokens · success) + evaluation gold-set harness
Mobile / wider reach	✓ **Expo mobile app** on a pnpm + Turborepo monorepo (shared client across web & mobile)
Content quality & safety	✓ **Off-syllabus quarantine** (34 docs retired) + **9-layer MCQ validator** and G1–G10 prompt guardrails

Grounded models over a curated corpus

Model chain — evolved in Phase 2

Text reasoning	OpenAI 4o-mini → Gemini 2.5-flash → Groq 70B
Quality paths	OpenAI gpt-5 (CA mocks, model answers)
OCR / vision	Gemini 2.0-flash → GPT-4o
Embeddings	text-embedding-3-small · 1536-d
Why	task-routing: cost · latency · no single-vendor risk

Data & retrieval

Sources	NCERTs, Laxmikanth, Spectrum, Shankar IAS, PYQs
Corpus	108 docs (74 in-syllabus) · 14,427 chunks
Question bank	36,658 generated · 26,901 live · 6,151 PYQ
Store	Supabase pgvector · LlamaIndex RAG
Knowledge graph	23,069 entities · 13,417 facts

RAG
grounds every question & citation

JSON
structured, parseable LLM output

No PII
anonymized concept IDs in all prompts

An eight-step evaluation pipeline

1 Auto-grade MCQs
flexible letter / text match · ⅓ negative marking

2 LLM-evaluate subjective
5-dimension rubric + directive verbs

3 Per-concept strength
& weakness derivation

4 Update mastery
SM-2 · 0.0–1.0 per concept

5 Write 4-tier memory
short / episodic / proc / semantic

6 Coverage records
per-chunk scoring

7 Revision schedule
spaced-repetition cards

8 Activity log
streaks & engagement

Instrumented

Every LLM call logs model, latency, tokens & success — **24,191 calls** captured for drift & cost tracking.

Latency SLO

Bounded component timeouts (RAG 8s · Exa 12s · LLM 20s); evaluation target **P50<8s, P95<15s**.

Resilient by design

Memory, coverage & activity writes are non-blocking — evaluation succeeds even if they fail.

How we test the AI — a representative gold set

1,580 expert-labelled items, stratified to mirror the live exam and content distribution.

EVAL SLICE	N	COMPOSITION	GROUND TRUTH
Prelims MCQ quality	**500**	7 subjects × 3 difficulties (20 / 45 / 35), proportional to the live 26,901-question bank	2 UPSC SMEs · $\kappa = 0.71$
RAG faithfulness	**300**	question + explanation pairs sampled across the 74 in-syllabus documents	LLM-judge + human spot-check
Mains subjective	**120**	GS1–GS4 + Essay + CA-linkage variant, spanning the full weak→excellent score range	3 expert evaluators (consensus)
Mistake classification	**200**	wrong answers labelled factual / conceptual / careless-slip / time-pressure	2 SMEs
Translation quality	**240**	30 items × 8 languages — text, options, explanation, model-answer sample	native bilingual reviewers
Handwriting OCR	**100**	multi-page Mains answer images, varied legibility & scripts	verified transcripts
Safety / adversarial	**120**	off-syllabus seeds, ambiguous MCQs, PII-injection, jailbreak / role-play	rule + human review

Why this set is defensible: stratified to the production mix (subject · difficulty · source · exam stage · 8 languages) so accuracy generalises; sized for ≤ ±3% at 95% CI on binary pass-rate; the Mains slice spans the full score range so correlation isn't inflated by the mid-band.

What the evaluation found

QUESTION GENERATION

Factual accuracy	96.4%
Single defensible answer	94.2%
In-syllabus relevance	99.2%
Statement specificity	92.0%
Exam-ready, no edits	89.2%

RAG GROUNDING

Faithfulness to source	95.7%
Citation accuracy	93.3%
Answerable from source	97.3%
Hallucinated citations	<2%

MAINS EVAL VS 3-EXPERT

Score MAE (/10)	0.61
Spearman ρ	0.84
Within ±1 band	91%
QWK vs experts	0.79

FACT-CHECK & MISTAKES

Contradiction precision	0.94
Contradiction recall	0.88
False docks (present.-only)	0
Mistake-class macro-F1	0.87

TRANSLATION (8 LANGS)

Adequacy (1–5)	4.5
Fluency (1–5)	4.3
Term preservation	99.3%
Script integrity	100%

HANDWRITING OCR

Character error rate	2.9%
Word error rate	11.8%
Legible-only CER	1.3%
Grade delta vs clean	±0.35

SAFETY / ROBUSTNESS

Off-syllabus rate	8.1→0.5%
PII in prompts	0
Jailbreak resistance	100%
Ambiguous-MCQ catch	94%

LATENCY / RELIABILITY*

Eval P50 / P95	6.2 / 10.8s
Fallback trigger rate	3.4%
Eval-survives-side-fail	100%
Translation cache hit	92%

*Latency / reliability measured from logged LLM calls. Quality figures are from the Phase-2 evaluation sprint over the 1,580-item gold set; every number here is reproducible via the **evals/** harness (re-run as a regression suite each release).

Tested with the aspirants we build for

WHO & HOW

- **12 representative users** — 5 repeat aspirants, 4 first-timers, 3 Hindi-preference; 8 from Tier-2/3 cities; budget-Android device mix. Recruited from the 96-user beta.
- **Moderated, think-aloud** task tests + System Usability Scale (SUS) + a feedback-trust rating.
- **4 tasks:** create + take an adaptive test · read the source-cited feedback · switch to Hindi & retake · act on the revision queue.

OUTCOMES

92%
task success rate

82
SUS score (excellent > 68)

4.5/5
feedback-trust rating

0.35→0.69
avg weak-concept mastery over 4 weeks

What we learned → shipped: (1) source citations were the #1 trust driver — kept front-and-centre; (2) "why this is wrong" was buried → moved above the fold; (3) Hindi users hit English strings → drove the **1,178-key** extraction across 16 catalogs; (4) "adaptive vs seed" confused users → default to adaptive with simpler copy.

User-validation figures are representative of the Phase-2 usability sprint with target-persona participants; engagement deltas drawn from beta cohort behaviour.

SYSTEM ARCHITECTURE

High-level architecture

CLIENTS	Web app · Next.js 16 + React		Mobile app · Expo shipped · pnpm + Turborepo monorepo	
API & ROUTING	FastAPI · Python 3.12 (async)		X-Lang locale seam + adaptive pool-first ~70% of questions served from cache, off the LLM	
AI ORCHESTRATION	**OpenAI → Gemini → Groq chain**	**RAG retriever · LlamaIndex**	**8-step eval + 4-tier memory**	**Translate-on-read + cache**
DATA & CONTENT	Supabase Postgres (v2)	pgvector embeddings	Upstash Redis cache	NewsPulse crawler Hindu · IE · Livemint · PIB

REAL-WORLD READINESS

Built for Bharat, not just metros

Regional languages — shipped

8 Indian languages (Hindi, Telugu, Tamil, Odia, Bengali, Marathi, Punjabi, Kannada). Questions, explanations & evaluation in the aspirant's own language — with UPSC terms, Articles & Acts preserved exactly.

Low connectivity

Text-first, lightweight payloads; adaptive pool-first keeps ~70% of actions off the LLM round-trip; evaluation can run deferred. Translations served from cache (92% warm-hit).

Entry-level devices

Web-first and light on the client, plus a native Expo app — designed to run on ₹6–8k Android phones with modest memory.

Offline use — in progress

Pre-generated question pools download as packs; attempts queue locally and sync — and evaluate — once a connection returns.

Cheap to run, cheaper to scale

Deployed on managed cloud

• LIVE

Status	Live in production · sambodh-ias.in
Web	**Vercel · Next.js 16**
API	**Render · FastAPI**
Database	**Supabase · pgvector**
Cache	**Upstash Redis**
Mobile	**Expo (managed)**
CI / CD	**GitHub Actions · auto-deploy from main**

SCALE	INFRA	LLM	TOTAL / MO	PER ACTIVE USER
Pilot ~150	$50	$25	**~$75**	fixed-cost dominated
1,000	$90	$150	**~$240**	**~₹20**
10,000	$300	$1,000	**~$1,300**	**~₹11**

Cost levers (in code today): adaptive pool-first ~70% off the LLM · translation cache 92% hit · Groq tier at **$0.00006/1K** tokens · prompt truncation ≤ 6K tokens · per-language model-answer cache.

₹ Marginal cost **~₹11–20 / active user / month** against **₹1–2 lakh / year** coaching — roughly **1/300th** the cost. A ₹199/month plan runs at **>90% gross margin**. Razorpay micro-unlocks (₹39–₹199) are already live.

Built to keep running — and to grow

MAINTENANCE — LOW-TOUCH BY DESIGN

- **Resilient:** 3-tier model fallback + feature flags; non-blocking side-writes mean an evaluation never fails because a memory/coverage write did.
- **Self-refreshing content:** NewsPulse crawler runs daily (6 AM IST) → monthly CA banks; off-syllabus quarantine is reversible.
- **Quality as regression:** dual-layer guardrails (prompt + validator); the eval gold set re-runs each release; 24,191 calls logged for drift.
- **Lean engineering:** live on fully-managed cloud (Vercel + Render + Supabase, auto-deploy from main) — no servers to babysit; monorepo + automated code review let one founder operate it.

ADOPTION PATH

- **Now (pilot):** 182 registered · 96 active · 148 evaluations · 12 early paying users on live ₹39–₹199 unlocks.
- **Wedge:** repeat aspirants in Tier-2/3 — freemium → Razorpay, the "desperate, specific" user coaching prices out.
- **Reach unlock (shipped):** 8 languages + mobile open the vernacular mass market English-only rivals can't serve.
- **Channels & ramp:** regional SEO, Telegram/YouTube cohorts, library & college tie-ins, Mains-2026 series hook → subscriptions + B2B2C.

North-star: concept gap-closure rate. **Moat:** vernacular reach × per-aspirant memory × a curated, syllabus-gated corpus — compounding with every attempt.

ROAD AHEAD · PHASE 3

From a Prelims + Mains engine to an AI-native institution

Harden

- Merge multilingual to production + admin translation review
- Offline / low-connectivity PWA to GA
- Auto-save & resume assessments

Deepen

- Peer benchmarking & comparative analytics
- Expand Mains corpus + Essay & Ethics depth
- Interview / Personality-test prep

Widen

- 3 more regional languages
- Same engine → State PSCs & other exams
- Community & P2P learning cohorts

🔗 Dependencies

Content digitization & licensing · OCR accuracy on handwriting · translation-quality review · LLM cost at scale.

🛡️ Risks & mitigation

Eval accuracy → RAG grounding + fact-check + human spot-checks. LLM cost → routing, caching, Groq tier. Scope → UPSC-first, adjacent exams only on the proven engine.

The team

Sabyasachi Upadhyay — Founder

Product Owner at Ola · Ex-NITI Aayog. Owns product, AI architecture, and the UPSC domain model end-to-end.

CORE TEAM

Priyajit Mohanty

Senior Manager, TD · Strategy & Mentor

Ayushi Agarwal

Ex-NITI Aayog · Quality Assurance (QA)

BUILDING OUT — KEY ROLES

AI / ML Engineer

RAG, evals, prompt & pipeline tuning

Backend Engineer

FastAPI, Postgres, async services

Frontend Engineer

Next.js, offline-first PWA

Content & SME

UPSC educators, rubric design

An AI-native institution for India's aspirants

Community & P2P learning

Peer cohorts and friendly competition to replace the isolation of solo preparation.

Multilingual access — live

8 Indian languages today, more on the roadmap — opening rigorous prep to millions more.

A new model of prep

Beyond UPSC to other competitive exams — AI-native, with offline support as the last resort.

APPENDIX

Adaptive selection algorithm

Question mix shifts with each learner's coverage — heavy on new ground early, heavy on correction and revision near mastery.

USER COVERAGE	CORRECTION (WEAK)	REVISION (MEDIUM)	NEW COVERAGE
0–1%	0%	0%	100%
1–30%	20%	10%	70%
30–70%	30%	20%	50%
70–95%	40%	30%	30%
95%+	50%	40%	10%

Per-chunk coverage status: **untested** (no attempts) · **weak** (<0.5) · **medium** (0.5–0.8) · **strong** (≥0.8).

APPENDIX

The 4-tier memory system

"Great test given!"

Short-term

Redis · 48h

Active session context and recent assessment data.

"I'm starting to understand you."

Episodic

Postgres · permanent

Assessment results and significant learning events.

"I know how you learn."

Procedural

Postgres · permanent

Study patterns, error patterns, learning strategies.

"I know who you are."

Semantic

Postgres · permanent

Concept cards, knowledge base, cached explanations.

Together they let the platform **distinguish temporary confusion from a fundamental gap** — so interventions match the real need.